

12.1 Orthogonal Functions

INTRODUCTION In certain areas of advanced mathematics, a function is considered to be a generalization of a vector. In this section we shall see how the two vector concepts of inner, or dot, product and orthogonality of vectors can be extended to functions. The remainder of the chapter is a practical application of this discussion.

Inner Product Recall, if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ are two vectors in R^3 or 3-space, then the inner product or dot product of $\mathbf{u}$ and $\mathbf{v}$ is a real number, called a **scalar**, defined as the sum of the products of their corresponding components:

In Chapter 7, the inner product was denoted by $\mathbf{u} \cdot \mathbf{v}$.

$$(\mathbf{u}, \mathbf{v}) = u_1v_1 + u_2v_2 + u_3v_3 = \sum_{k=1}^3 u_kv_k.$$

The inner product $(\mathbf{u}, \mathbf{v})$ possesses the following properties:

- (i) $(\mathbf{u}, \mathbf{v}) = (\mathbf{v}, \mathbf{u})$
- (ii) $(k\mathbf{u}, \mathbf{v}) = k(\mathbf{u}, \mathbf{v})$, k a scalar
- (iii) $(\mathbf{u}, \mathbf{u}) = 0$ if $\mathbf{u} = \mathbf{0}$ and $(\mathbf{u}, \mathbf{u}) > 0$ if $\mathbf{u} \neq \mathbf{0}$
- (iv) $(\mathbf{u} + \mathbf{v}, \mathbf{w}) = (\mathbf{u}, \mathbf{w}) + (\mathbf{v}, \mathbf{w})$.

We expect any generalization of the inner product to possess these same properties.

Suppose that f_1 and f_2 are piecewise-continuous functions defined on an interval $[a, b]$.* Since a definite integral on the interval of the product $f_1(x)f_2(x)$ possesses properties (i)–(iv) of the inner product of vectors, whenever the integral exists we are prompted to make the following definition.

Definition 12.1.1 Inner Product of Functions

The **inner product** of two functions f_1 and f_2 on an interval $[a, b]$ is the number

$$(f_1, f_2) = \int_a^b f_1(x)f_2(x) dx.$$

Orthogonal Functions Motivated by the fact that two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal whenever their inner product is zero, we define **orthogonal functions** in a similar manner.

Definition 12.1.2 Orthogonal Functions

Two functions f_1 and f_2 are said to be **orthogonal** on an interval $[a, b]$ if

$$(f_1, f_2) = \int_a^b f_1(x)f_2(x) dx = 0. \quad (1)$$

EXAMPLE 1 Orthogonal Functions

The functions $f_1(x) = x^2$ and $f_2(x) = x^3$ are orthogonal on the interval $[-1, 1]$. This fact follows from (1):

$$(f_1, f_2) = \int_{-1}^1 x^2 \cdot x^3 dx = \int_{-1}^1 x^5 dx = \left[\frac{1}{6} x^6 \right]_{-1}^1 = 0. \quad \equiv$$

Unlike vector analysis, where the word *orthogonal* is a synonym for *perpendicular*, in this present context the term *orthogonal* and condition (1) have no geometric significance.

Orthogonal Sets We are primarily interested in infinite sets of orthogonal functions.

*The interval could also be $(-\infty, \infty)$, $[0, \infty)$, and so on.

Definition 12.1.3 Orthogonal Set

A set of real-valued functions $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is said to be **orthogonal** on an interval $[a, b]$ if

$$(\phi_m, \phi_n) = \int_a^b \phi_m(x) \phi_n(x) dx = 0, \quad m \neq n. \quad (2)$$

Orthonormal Sets The norm, or length $\|\mathbf{u}\|$, of a vector $\mathbf{u}$ can be expressed in terms of the inner product. The expression $(\mathbf{u}, \mathbf{u}) = \|\mathbf{u}\|^2$ is called the square norm, and so the norm is $\|\mathbf{u}\| = \sqrt{(\mathbf{u}, \mathbf{u})}$. Similarly, the **square norm** of a function ϕ_n is $\|\phi_n(x)\|^2 = (\phi_n, \phi_n)$, and so the **norm**, or its generalized length, is $\|\phi_n(x)\| = \sqrt{(\phi_n, \phi_n)}$. In other words, the square norm and norm of a function ϕ_n in an orthogonal set $\{\phi_n(x)\}$ are, respectively,

$$\|\phi_n(x)\|^2 = \int_a^b \phi_n^2(x) dx \quad \text{and} \quad \|\phi_n(x)\| = \sqrt{\int_a^b \phi_n^2(x) dx}. \quad (3)$$

If $\{\phi_n(x)\}$ is an orthogonal set of functions on the interval $[a, b]$ with the property that $\|\phi_n(x)\| = 1$ for $n = 0, 1, 2, \dots$, then $\{\phi_n(x)\}$ is said to be an **orthonormal set** on the interval.

EXAMPLE 2 Orthogonal Set of Functions

Show that the set $\{1, \cos x, \cos 2x, \dots\}$ is orthogonal on the interval $[-\pi, \pi]$.

SOLUTION If we make the identification $\phi_0(x) = 1$ and $\phi_n(x) = \cos nx$, we must then show that $\int_{-\pi}^{\pi} \phi_0(x) \phi_n(x) dx = 0$, $n \neq 0$, and $\int_{-\pi}^{\pi} \phi_m(x) \phi_n(x) dx = 0$, $m \neq n$. We have, in the first case, for $n \neq 0$,

$$\begin{aligned} (\phi_0, \phi_n) &= \int_{-\pi}^{\pi} \phi_0(x) \phi_n(x) dx = \int_{-\pi}^{\pi} \cos nx dx \\ &= \frac{1}{n} \sin nx \Big|_{-\pi}^{\pi} = \frac{1}{n} [\sin n\pi - \sin(-n\pi)] = 0, \end{aligned}$$

and in the second, for $m \neq n$,

$$\begin{aligned} (\phi_m, \phi_n) &= \int_{-\pi}^{\pi} \phi_m(x) \phi_n(x) dx = \int_{-\pi}^{\pi} \cos mx \cos nx dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} [\cos(m+n)x + \cos(m-n)x] dx \quad \leftarrow \text{trigonometric identity} \\ &= \frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right]_{-\pi}^{\pi} = 0. \end{aligned} \quad \equiv$$

EXAMPLE 3 Norms

Find the norms of each function in the orthogonal set given in Example 2.

SOLUTION For $\phi_0(x) = 1$ we have from (3)

$$\|\phi_0(x)\|^2 = \int_{-\pi}^{\pi} dx = 2\pi$$

so that $\|\phi_0(x)\| = \sqrt{2\pi}$. For $\phi_n(x) = \cos nx$, $n > 0$, it follows that

$$\|\phi_n(x)\|^2 = \int_{-\pi}^{\pi} \cos^2 nx dx = \frac{1}{2} \int_{-\pi}^{\pi} [1 + \cos 2nx] dx = \pi.$$

Thus for $n > 0$, $\|\phi_n(x)\| = \sqrt{\pi}$. ≡

An orthogonal set can be made into an orthonormal set. 

Any orthogonal set of nonzero functions $\{\phi_n(x)\}$, $n = 0, 1, 2, \dots$, can be *normalized*—that is, made into an orthonormal set—by dividing each function by its norm. It follows from Examples 2 and 3 that the set

$$\left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \dots \right\}$$

is orthonormal on the interval $[-\pi, \pi]$.

Vector Analogy We shall make one more analogy between vectors and functions. Suppose $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ are three mutually orthogonal nonzero vectors in 3-space. Such an orthogonal set can be used as a basis for 3-space; that is, any three-dimensional vector can be written as a linear combination

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3, \quad (4)$$

where the c_i , $i = 1, 2, 3$, are scalars called the components of the vector. Each component c_i can be expressed in terms of $\mathbf{u}$ and the corresponding vector $\mathbf{v}_i$. To see this we take the inner product of (4) with $\mathbf{v}_1$:

$$(\mathbf{u}, \mathbf{v}_1) = c_1(\mathbf{v}_1, \mathbf{v}_1) + c_2(\mathbf{v}_2, \mathbf{v}_1) + c_3(\mathbf{v}_3, \mathbf{v}_1) = c_1\|\mathbf{v}_1\|^2 + c_2 \cdot 0 + c_3 \cdot 0.$$

Hence

$$c_1 = \frac{(\mathbf{u}, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}.$$

In like manner we find that the components c_2 and c_3 are given by

$$c_2 = \frac{(\mathbf{u}, \mathbf{v}_2)}{\|\mathbf{v}_2\|^2} \quad \text{and} \quad c_3 = \frac{(\mathbf{u}, \mathbf{v}_3)}{\|\mathbf{v}_3\|^2}.$$

Hence (4) can be expressed as

$$\mathbf{u} = \frac{(\mathbf{u}, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{(\mathbf{u}, \mathbf{v}_2)}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \frac{(\mathbf{u}, \mathbf{v}_3)}{\|\mathbf{v}_3\|^2} \mathbf{v}_3 = \sum_{n=1}^3 \frac{(\mathbf{u}, \mathbf{v}_n)}{\|\mathbf{v}_n\|^2} \mathbf{v}_n. \quad (5)$$

Orthogonal Series Expansion Suppose $\{\phi_n(x)\}$ is an infinite orthogonal set of functions on an interval $[a, b]$. We ask: If $y = f(x)$ is a function defined on the interval $[a, b]$, is it possible to determine a set of coefficients c_n , $n = 0, 1, 2, \dots$, for which

$$f(x) = c_0\phi_0(x) + c_1\phi_1(x) + \dots + c_n\phi_n(x) + \dots? \quad (6)$$

As in the foregoing discussion on finding components of a vector, we can find the coefficients c_n by utilizing the inner product. Multiplying (6) by $\phi_m(x)$ and integrating over the interval $[a, b]$ gives

$$\begin{aligned} \int_a^b f(x)\phi_m(x) dx &= c_0 \int_a^b \phi_0(x)\phi_m(x) dx + c_1 \int_a^b \phi_1(x)\phi_m(x) dx + \dots + c_n \int_a^b \phi_n(x)\phi_m(x) dx + \dots \\ &= c_0(\phi_0, \phi_m) + c_1(\phi_1, \phi_m) + \dots + c_n(\phi_n, \phi_m) + \dots \end{aligned}$$

By orthogonality, each term on the right-hand side of the last equation is zero *except* when $m = n$. In this case we have

$$\int_a^b f(x)\phi_n(x) dx = c_n \int_a^b \phi_n^2(x) dx.$$

It follows that the required coefficients c_n are given by

$$c_n = \frac{\int_a^b f(x)\phi_n(x) dx}{\int_a^b \phi_n^2(x) dx}, \quad n = 0, 1, 2, \dots$$

In other words,
$$f(x) = \sum_{n=0}^{\infty} c_n \phi_n(x), \quad (7)$$

where
$$c_n = \frac{\int_a^b f(x) \phi_n(x) dx}{\|\phi_n(x)\|^2}. \quad (8)$$

With inner product notation, (7) becomes

$$f(x) = \sum_{n=0}^{\infty} \frac{(f, \phi_n)}{\|\phi_n(x)\|^2} \phi_n(x). \quad (9)$$

Thus (9) is seen to be the function analogue of the vector result given in (5).

Definition 12.1.4 Orthogonal Set/Weight Function

A set of real-valued functions $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is said to be **orthogonal with respect to a weight function** $w(x)$ on an interval $[a, b]$ if

$$\int_a^b w(x) \phi_m(x) \phi_n(x) dx = 0, \quad m \neq n.$$

The usual assumption is that $w(x) > 0$ on the interval of orthogonality $[a, b]$. The set $\{1, \cos x, \cos 2x, \dots\}$ in Example 2 is orthogonal with respect to the weight function $w(x) = 1$ on the interval $[-\pi, \pi]$.

If $\{\phi_n(x)\}$ is orthogonal with respect to a weight function $w(x)$ on the interval $[a, b]$, then multiplying (6) by $w(x)\phi_n(x)$ and integrating yields

$$c_n = \frac{\int_a^b f(x) w(x) \phi_n(x) dx}{\|\phi_n(x)\|^2}, \quad (10)$$

where
$$\|\phi_n(x)\|^2 = \int_a^b w(x) \phi_n^2(x) dx. \quad (11)$$

The series (7) with coefficients c_n given by either (8) or (10) is said to be an **orthogonal series expansion** of f or a **generalized Fourier series**.

Complete Sets The procedure outlined for determining the coefficients c_n was *formal*; that is, basic questions on whether an orthogonal series expansion such as (7) is actually possible were ignored. Also, to expand f in a series of orthogonal functions, it is certainly necessary that f not be orthogonal to each ϕ_n of the orthogonal set $\{\phi_n(x)\}$. (If f were orthogonal to every ϕ_n , then $c_n = 0$, $n = 0, 1, 2, \dots$) To avoid the latter problem we shall assume, for the remainder of the discussion, that an orthogonal set is **complete**. This means that the only continuous function orthogonal to each member of the set is the zero function.

REMARKS

Suppose that $\{f_0(x), f_1(x), f_2(x), \dots\}$ is an infinite set of real-valued functions that are continuous on an interval $[a, b]$. If this set is *linearly independent* on $[a, b]$ (see page 357 for the definition of an infinite linearly independent set), then it can always be made into an orthogonal set and, as described earlier in this section, can be made into an orthonormal set. See Problem 27 in Exercises 12.1.

In Problems 1–6, show that the given functions are orthogonal on the indicated interval.

1. $f_1(x) = x, f_2(x) = x^2; \quad [-2, 2]$
2. $f_1(x) = x^3, f_2(x) = x^2 + 1; \quad [-1, 1]$
3. $f_1(x) = e^x, f_2(x) = xe^{-x} - e^{-x}; \quad [0, 2]$
4. $f_1(x) = \cos x, f_2(x) = \sin^2 x; \quad [0, \pi]$
5. $f_1(x) = x, f_2(x) = \cos 2x; \quad [-\pi/2, \pi/2]$
6. $f_1(x) = e^x, f_2(x) = \sin x; \quad [\pi/4, 5\pi/4]$

In Problems 7–12, show that the given set of functions is orthogonal on the indicated interval. Find the norm of each function in the set.

7. $\{\sin x, \sin 3x, \sin 5x, \dots\}; \quad [0, \pi/2]$
8. $\{\cos x, \cos 3x, \cos 5x, \dots\}; \quad [0, \pi/2]$
9. $\{\sin nx\}, \quad n = 1, 2, 3, \dots; \quad [0, \pi]$
10. $\left\{\sin \frac{n\pi}{p}x\right\}, \quad n = 1, 2, 3, \dots; \quad [0, p]$
11. $\left\{1, \cos \frac{n\pi}{p}x\right\}, \quad n = 1, 2, 3, \dots; \quad [0, p]$
12. $\left\{1, \cos \frac{n\pi}{p}x, \sin \frac{m\pi}{p}x\right\}, \quad n = 1, 2, 3, \dots, \\ m = 1, 2, 3, \dots; \quad [-p, p]$

In Problems 13 and 14, verify by direct integration that the functions are orthogonal with respect to the indicated weight function on the given interval.

13. $H_0(x) = 1, H_1(x) = 2x, H_2(x) = 4x^2 - 2; \quad w(x) = e^{-x^2}, \quad (-\infty, \infty)$
14. $L_0(x) = 1, L_1(x) = -x + 1, L_2(x) = \frac{1}{2}x^2 - 2x + 1; \quad w(x) = e^{-x}, \quad [0, \infty)$
15. Let $\{\phi_n(x)\}$ be an orthogonal set of functions on $[a, b]$ such that $\phi_0(x) = 1$. Show that $\int_a^b \phi_n(x) dx = 0$ for $n = 1, 2, \dots$
16. Let $\{\phi_n(x)\}$ be an orthogonal set of functions on $[a, b]$ such that $\phi_0(x) = 1$ and $\phi_1(x) = x$. Show that $\int_a^b (\alpha x + \beta)\phi_n(x) dx = 0$ for $n = 2, 3, \dots$ and any constants α and β .
17. Let $\{\phi_n(x)\}$ be an orthogonal set of functions on $[a, b]$. Show that $\|\phi_m(x) + \phi_n(x)\|^2 = \|\phi_m(x)\|^2 + \|\phi_n(x)\|^2, m \neq n$.
18. From Problem 1 we know that $f_1(x) = x$ and $f_2(x) = x^2$ are orthogonal on $[-2, 2]$. Find constants c_1 and c_2 such that $f_3(x) = x + c_1x^2 + c_2x^3$ is orthogonal to both f_1 and f_2 on the same interval.
19. The set of functions $\{\sin nx\}, n = 1, 2, 3, \dots$, is orthogonal on the interval $[-\pi, \pi]$. Show that the set is not complete.
20. Suppose f_1, f_2 , and f_3 are functions continuous on the interval $[a, b]$. Show that $(f_1 + f_2, f_3) = (f_1, f_3) + (f_2, f_3)$.

A real-valued function is said to be **periodic** with period $T \neq 0$ if $f(x + T) = f(x)$ for all x in the domain of f . If T is the smallest positive value for which $f(x + T) = f(x)$ holds, then T is called the **fundamental period** of f . In Problems 21–26, determine the fundamental period T of the given function.

21. $f(x) = \cos 2\pi x$
22. $f(x) = \sin \frac{4}{L}x, L > 0$
23. $f(x) = \sin x + \sin 2x$
24. $f(x) = \sin 2x + \cos 4x$
25. $f(x) = \sin 3x + \cos 2x$
26. $f(x) = \sin^2 \pi x$

Discussion Problems

27. The **Gram–Schmidt process** for constructing an orthogonal set that was discussed in Section 7.7 carries over to a linearly independent set $\{f_0(x), f_1(x), f_2(x), \dots\}$ of real-valued functions continuous on an interval $[a, b]$. With the inner product $(f_n, \phi_n) = \int_a^b f_n(x)\phi_n(x) dx$, define the functions in the set $B' = \{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ to be

$$\phi_0(x) = f_0(x)$$

$$\phi_1(x) = f_1(x) - \frac{(f_1, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x)$$

$$\phi_2(x) = f_2(x) - \frac{(f_2, \phi_0)}{(\phi_0, \phi_0)} \phi_0(x) - \frac{(f_2, \phi_1)}{(\phi_1, \phi_1)} \phi_1(x)$$

$$\vdots \qquad \qquad \qquad \vdots$$

and so on.

- (a) Write out $\phi_3(x)$ in the set.
- (b) By construction, the set $B' = \{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is orthogonal on $[a, b]$. Demonstrate that $\phi_0(x), \phi_1(x)$, and $\phi_2(x)$ are mutually orthogonal.
28. (a) Consider the set of functions $\{1, x, x^2, x^3, \dots\}$ defined on the interval $[-1, 1]$. Apply the Gram–Schmidt process given in Problem 27 to this set and find $\phi_0(x), \phi_1(x), \phi_2(x)$, and $\phi_3(x)$ of the orthogonal set B' .
- (b) Discuss: Do you recognize the orthogonal set?
29. Verify that the inner product (f_1, f_2) in Definition 12.1.1 satisfies properties (i)–(iv) given on page 672.
30. In R^3 , give an example of a set of orthogonal vectors that is not complete. Give a set of orthogonal vectors that is complete.
31. The function

$$f(x) = A_0 + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi}{p}x + B_n \sin \frac{n\pi}{p}x \right),$$

where the coefficients A_n and B_n depend only on n , is periodic. Find the period T of f .